The goal for this study is to determine the optimal price for a business, based on a range between $0 and $70. With the different choices that customers make for a given price, the study uses simulations to determine how demand responds to price. At the optimum, each consumer should have a profit of $11.925.

My design handles uncertainty about changes in revenue by using a sales function. The sales function has an argument with mean customers that allows for easier control over the number of customers in the experiment. It also has an rng argument that allows for random number generation. There is also a price parameter that returns a vector of customer choices with mean customers. It uses stochastic optimization to set each price as an experiment with information. Revenue is found from customer observations based on price changes rather than a fixed outcome.

My design selects prices by using local search to combine inference and optimization. A slope is estimated to find the price. The search uses several prices across the range. Through this, the price in the neighborhood is refined with fresh customer data and the price with best validation revenue is selected.

One challenge I faced was that random variation makes it hard to tell if change in revenue comes from price or a random confounding variable. I addressed this by comparing revue at nearby prices through repeated simulations. I also used standard errors to interpret the uncertainty of these comparisons.

Another challenge I faced was adapting the code used in the local search example for the lab with the gradient ascent only being a local search rather than global. I addressed this by running eight starting prices (0, 10, 20, 30, 40, 50, 60, 70). The results were compared with

separate validation samples rather than purely search data. This also makes it less likely that the starting prices affected the results.

For my final results, the optimal price was $27.38. There are 483985 consumers experimented on for this study. The revenue during the experiment was $11.885. The study captured 99.7% of the optimal profit.